The Geometric Boundary of Few-Shot Rule Induction in Frontier LLMs

Spatial Separability, not Output Cardinality, Predicts In-Context Recovery

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Abstract

We test whether the geometric structure of a parameterized rule predicts whether commercial frontier-family large language models will recover it from $K{=}16$ in-context input–output examples. We design a cardinality-controlled falsification experiment with three independent task-suite seeds: at output cardinality 4, spatially separable rules ($y = \lfloor x/50 \rfloor$) are recovered at a mean accuracy of 0.88 across three commercial frontier-family models (Anthropic Haiku 4.5, Sonnet 4.6, OpenAI GPT-4o-mini), while algebraically coupled rules with the same cardinality ($y = ((ax + c) \bmod m) \bmod 4$) fall to 0.25 ($\Delta = 0.63 \pm 0.02$ across seeds). At cardinality 50, the same comparison gives 0.88 versus 0.02 ($\Delta = 0.86 \pm 0.01$). By-task permutation tests yield $p < 10^{-4}$ on all six per-model contrasts. Best-of-3 ensemble accuracy reaches 1.00 on spatial families at both cardinalities and only 0.56 ± 0.03 and 0.05 ± 0.03 respectively on algebraic families: pooling LLMs does not cross the boundary. The cardinality hypothesis ("larger output space is harder") predicts the opposite of what we observe and is falsified. We discuss implications for pre-deployment LLM capability prediction, cost-optimized routing, and symbolic-fallback hybrid pipelines, and frame the result as the empirical analogue at the frontier-LLM level of the architecture-level distinction in Dovzak (2026).

1 Introduction

A common engineering question in deployments of LLM-driven agents is whether a given rule, expressed by K input–output examples shown in-context, will be reliably recovered by the language model and applied to new inputs. Existing predictors of LLM downstream performance fall into two categories: scaling-law predictors that use compute and pre-training loss as inputs [2, 5, 3], and task-graph predictors that use compositional depth or width [4]. Neither category asks whether the *geometric structure* of the rule itself, computable from the task specification without invoking the LLM, predicts induction success.

We make three contributions:

1. **A clean empirical partition.** Across three commercial frontier-family LLMs from two vendors, accuracy on $K{=}16$ in-context induction partitions sharply along an axis we call *spatial separability versus algebraic coupling.* The gap is consistent across vendors and across two output-cardinality regimes.
2. **A cardinality-controlled falsification test.** We construct a 40-task suite in which output cardinality is matched between spatially separable and algebraically coupled families. The geometric gap survives. The naive null hypothesis "larger output space is harder" is reversed: at cardinality 50, spatial accuracy is at ceiling while algebraic accuracy is at floor.

3. **A best-of-N analysis.** Pooling three commercial frontier-family LLMs as an ensemble lifts spatial accuracy to 1.00 at both tested cardinalities but does not push algebraic accuracy past 0.56 ± 0.03 at cardinality 4 or 0.05 ± 0.03 at cardinality 50 across seeds. The boundary is not a per-model artifact; it is a property of the rule space.

We position this paper as the empirical follow-up to [1], which established the same geometric distinction at the architecture level for purpose-built K-shot induction networks. The present paper asks whether the boundary persists when the inducer is a frontier LLM. The answer is yes, and the boundary is sharper.

2 Related work

Predicting LLM downstream performance. Scaling-law approaches predict accuracy from compute, parameter count, and pre-training loss [5, 2]. [3] argues that emergent capabilities are an artifact of nonlinear metrics rather than fundamental capability shifts. [4] predicts failure on multi-step compositional problems using task-graph properties (depth, width). Neither uses the geometric structure of the rule itself as the predictor variable, and neither asks whether classes of rules with similar task-graph depth differ in solvability based on parameter coupling.

In-context learning of operators. [6] establishes that transformers can implement linear regression in-context. [7] characterizes the in-context-learning regime as implicit gradient descent on a linear hypothesis. The DeepOSets line [8, 9] establishes permutation-invariant operator induction in the continuous regime. None of these works names a structural boundary or pairs a clean success regime with a clean failure regime under matched conditions.

Hypothesis-driven LLM rule induction. [10] prompts an LLM to propose natural-language hypotheses, implements them as Python programs, and verifies on examples; achieves 30% on a 100-problem ARC subset versus 17% direct prompting. The pipeline is orthogonal to our axis: their improvement is on hypothesis-search efficiency, not on which task families the underlying LLM can recover from $K = 16$ examples directly.

Trained vs in-context modular arithmetic. A separate line of work shows that transformers *trained* on modular addition or multiplication eventually grok the operation and implement it via Fourier-basis circuits [15] or irreducible group representations [16]. [17] trains transformers from scratch on linear congruential generator sequences and finds they develop digit-wise factorization circuits, eventually generalizing to unseen moduli up to $m_{\text{test}} = 2^{16}$ via a two-step "estimate m then factorize" strategy. The mechanism, when it succeeds, is a learned circuit. [18] characterizes the geometric structure of transformer learning trajectories on modular arithmetic and finds that effective execution dimension scales with task complexity, suggesting task-structure predictors are extractable from training dynamics. Our finding does not contradict any of these. We claim that frontier LLMs do not recover algebraic operators from K=16 *in-context examples* at inference time, with no parameter update. The trained-circuit and in-context-recovery regimes appear distinct: the first is a stored procedure built up over training, the second would require either inferring the operator from ~ 16 examples or having pre-trained on the specific (a, c, m), both of which our evidence indicates frontier LLMs are not doing.

LLM self-prediction of capability. [19] asks whether LLMs can predict their own success on a task before attempting it. The framing is different from ours: in their setup, the LLM is the predictor; in ours, the predictor is an external classifier on geometric features computable without invoking any LLM. They report that LLMs are systematically overconfident. Our predictor is offline (no LLM call), which both bounds cost and removes the overconfidence failure mode.

Compositional generalization. [11] taxonomy and follow-ups [12, 13] characterize compositional gaps along length, structure, and lexical dimensions. The boundary we describe is orthogonal: it is about parameter identifiability rather than depth or length.

Local identifiability. Our formal condition follows the local-identifiability tradition of [14]: full rank of the parameter-to-output Jacobian over a sub-region of the support. We use this language because it gives a precise verbal name to a property that is otherwise typically left implicit in the few-shot induction literature.

3 Setup

3.1 K-shot operator induction

A K-shot operator induction episode is a tuple $(\mathcal{F}, \theta, S, q)$ where $\mathcal{F}$ is a parametric operator family, $\theta \in \Theta$ is the per-episode parameter vector, $S = \{(x_i, y_i = \mathcal{F}_\theta(x_i))\}_{i=1}^{K}$ is the support, and q is a query input not in S. The model receives $(\mathcal{F}, S, q)$ as a prompt and is asked to output the predicted $\hat{y}$. In every experiment in this paper, $K = 16$.

3.2 Spatial separability

Let $J_\theta(x) = \partial \mathcal{F}_\theta(x)/\partial\theta$ be the Jacobian of $\mathcal{F}_\theta$ with respect to θ at input x. We say $\mathcal{F}$ admits *spatially separable parameters* over a support distribution $\mathcal{D}$ if for every parameter coordinate θ_k there exists a measurable sub-region $R_k \subseteq \mathrm{supp}(\mathcal{D})$ such that $J_\theta(x)$ restricted to R_k has rank one with the dominant direction along θ_k, and $\Pr_{x\sim\mathcal{D}}(x \in R_k) > 0$.

Concretely: there is some sub-region of the input space where each parameter "shows up alone." This is local identifiability in the sense of [14]. Threshold $y = \mathbb{1}[x > T]$ is spatially separable: the threshold T is identifiable from the value of x alone; sub-regions $\{x : x > T\}$ and $\{x : x \leq T\}$ are sufficient for unique identification. Conjunctive thresholds $z = \mathbb{1}[x > T_1 \wedge y > T_2]$ are spatially separable: T_1 is identifiable from samples with y fixed, T_2 symmetrically.

A note on formality. The Jacobian-based characterization above is clean for smooth parametric families but is, strictly, slippery for discrete and modular operators where the partial derivatives are zero almost everywhere. We use this language as an operational geometric distinction rather than a complete formal taxonomy. The five families we evaluate fall on opposite sides of an empirically clean partition; the formal extension of local identifiability to discrete-modular settings is open.

3.3 Algebraic coupling

We say $\mathcal{F}$ has *algebraically coupled parameters* over $\mathcal{D}$ if the Jacobian columns $\partial\mathcal{F}_\theta(x)/\partial\theta_k$ are linearly dependent for every $x \in \mathrm{supp}(\mathcal{D})$, so no parameter is locally privileged, and the parameters are entangled through a shared equation. Examples: linear congruential map $f(x) = (ax + c) \bmod m$ in the parameters (a, c, m), modular polynomial $f(x) = (ax^2 + bx + c) \bmod p$ in (a, b, c) with p given. Recovering any single parameter requires jointly fitting all of them.

3.4 Hypothesis

> *Frontier large language models recover rules whose parameters are spatially separable, at K=16 examples, with substantially higher accuracy than rules whose parameters are algebraically coupled, under matched output cardinality and matched input distribution.*

The matched-cardinality clause is essential: spatially separable benchmark rules tend to have small output spaces (e.g. binary thresholds), and a naive analysis would conflate "spatial" with "small output space." We design Section 5 explicitly to disambiguate these axes.

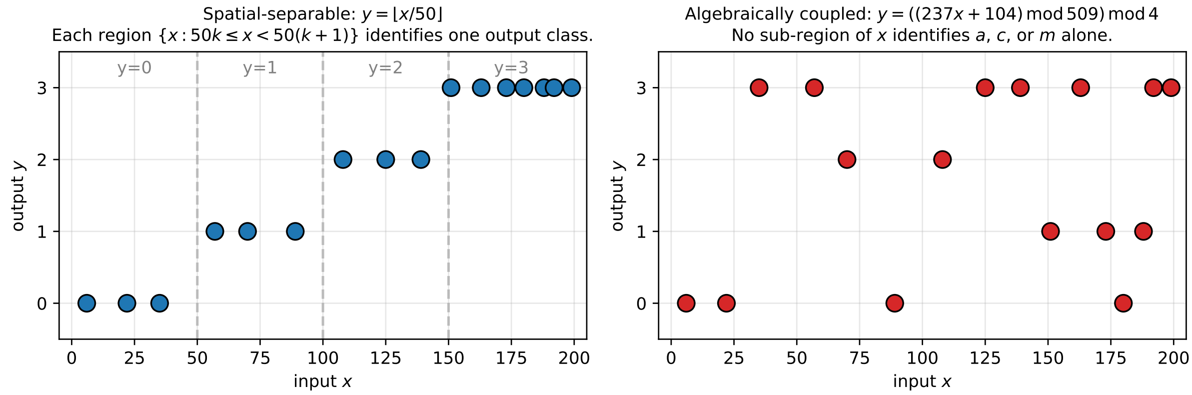

Figure 1: Conceptual contrast between the two regimes at matched output cardinality (4). **Left:** a spatial-separable rule, $y = \lfloor x/50 \rfloor$. The output partitions x into four contiguous regions; the threshold parameters are locally identifiable from any region. **Right:** an algebraically coupled rule, $y = ((237x + 104) \bmod 509) \bmod 4$, with the same domain and the same 4 output classes. The map from x to y is deterministic but no sub-region of x identifies a, c, or m alone; recovery requires joint identification across the full support.

3.5 Models

We evaluate three commercial frontier-family LLMs accessed via vendor APIs:

- Anthropic Claude Haiku 4.5 (`claude-haiku-4-5-20251001`)
- Anthropic Claude Sonnet 4.6 (`claude-sonnet-4-6`)
- OpenAI GPT-4o-mini (`gpt-4o-mini`)

Anthropic Claude Opus 4.7 (`claude-opus-4-7`) is also evaluated on the pilot suite of Section 4; it is omitted from the cardinality test for cost. Open-weight frontier models are not evaluated; we discuss this as a limitation in Section 9.

3.6 Prompt protocol and parsing

Each query is a fresh API call with no shared session. The system message frames the task as deterministic rule induction and instructs chain-of-thought followed by a structured final answer. The user message provides the K=16 examples and the query in canonical text form (`x = 174 -> y = 1`, one example per line) and ends with the instruction

```
After your reasoning, write your answer on a SEPARATE FINAL LINE in this exact
format:
FINAL ANSWER: <integer>
```

Maximum output tokens is 2000. Outputs are parsed by regex matching `FINAL ANSWER: <int>` as the primary pattern, with "answer is X" and last-integer-in-response as fallback. In the cardinality experiment, 597 of 600 (99.5%) responses parsed via the primary pattern.

3.7 Validation protocol

We use four robustness checks:

- *Bootstrap by-task confidence intervals*: 2000 resamples drawn at the task level (not the query level), to respect the fact that 5 queries from the same task share a parameter θ and are therefore not independent.
- *By-task permutation tests*: 5000 permutations of task–family labels, computing observed Δ and p-value.
- *Per-task variance audit*: per-task accuracy distribution to ensure no outlier task drives the cell mean.
- *Hand audit*: 30 random rows manually verified for parser correctness against the raw model output.

4 Pilot: tasks across five operator families

We first measure accuracy on five operator families to confirm the gap exists in a baseline regime before designing the cardinality test. Per-family details are in Appendix A. We use 50 tasks per family (10 tasks $\times$ 5 queries each = 50 task-queries per cell).

Table 1: Pilot suite: per-model per-family accuracy. Each cell is mean over 50 task-queries (10 tasks $\times$5 queries). Sonnet uses the chain-of-thought protocol of Section 3.6; Haiku, Opus, and GPT-4o-mini use a terse-prompt protocol with $\max\_tokens = 15$ in this pilot. Section 5 re-runs all models under the unified protocol; numbers there should be considered the methodologically clean reference.

Family	Geometry	Haiku 4.5	Sonnet 4.6	Opus 4.7	GPT-4o-mini
threshold	spatial	0.74	0.50	0.88	0.60
conjunctive	spatial	0.66	0.10	0.68	0.62
decision_tree	spatial	0.68	0.12	0.10	0.36
lcg	algebraic	0.18	0.20	0.20	0.20
polycoef	algebraic	0.12	0.02	0.12	0.10
spatial mean		0.69	0.24	0.55	0.53
algebraic mean		0.15	0.11	0.16	0.15
gap		**+0.54**	**+0.13**	**+0.39**	**+0.38**

The per-model gap ranges from +0.13 (Sonnet 4.6) to +0.54 (Haiku 4.5). Three of four models exhibit a gap $\geq +0.38$. The Sonnet 4.6 gap is small in this pilot; in Section 5, under the unified protocol, Sonnet's spatial accuracy reaches the ceiling and the gap recovers to +0.66 to +0.96.

The pilot has two limitations that motivate the cardinality experiment: (i) protocol heterogeneity across models, and (ii) confound between geometry and output cardinality (spatial families have 2–4 output classes, algebraic families have $\sim$50–200). **The methodologically clean reference for our central claim is Section 5; the pilot exists only to motivate the design of the cardinality experiment.**

5 Cardinality-controlled falsification test

5.1 Design

We construct four families that disentangle geometry from output cardinality. All four use 1-D integer inputs $x \in [0, 199]$, $K = 16$ in-context examples, and 5 held-out queries per task.

Family	Geometry	Output cardinality
`bucket_4_spatial`	spatial	4 ($y = \lfloor x/50 \rfloor$)
`bucket_50_spatial`	spatial	50 ($y = \lfloor x/4 \rfloor$)
`lcg_mod4_algeb`	algebraic	4 ($y = ((ax + c) \bmod m) \bmod 4$)
`lcg_mod50_algeb`	algebraic	50 ($y = ((ax + c) \bmod m) \bmod 50$)

The `bucket_*` families are pure 1D spatial bucketing: each output bucket corresponds to a contiguous interval of x. The `lcg_mod*` families take the full LCG output and reduce modulo the target cardinality, so they are algebraically coupled (require recovery of a, c, m) at the chosen output cardinality. We use prime $m \in [503, 1009]$ so that $K = 16$ examples cover at most 3.18% of the LCG cycle, ruling out cycle memorization. For the cardinality-4 families (`bucket_4_spatial` and `lcg_mod4_algeb`), output-class coverage is enforced by rejection sampling so that all 4 output classes appear in the K=16 in-context examples. For the cardinality-50 families, we require at least 10 of the 50 output classes to be observed in the support (verified empirically: 12 to 16 classes per task, mean 13.8).

The pre-registered prediction:

- If output cardinality drives accuracy: acc(bucket_4) $\approx$ acc(lcg_mod4) and acc(bucket_50) $\approx$ acc(lcg_mod50), both pairs falling with cardinality.
- If geometry drives accuracy: acc(bucket_*) $\gg$ acc(lcg_mod*) at every cardinality.

We run three commercial frontier-family LLMs (Haiku 4.5, Sonnet 4.6, GPT-4o-mini) on 40 tasks (10 per family) with 5 queries per task per model, repeated independently across 3 task-suite seeds (42, 17, 1337): 1,800 API calls per model, 5,400 total. Total API cost across the three seeds was approximately $12.

5.2 Results

Table 2: Cardinality-controlled accuracy across three independent task-suite seeds. Cells are mean ± std over seeds (each seed: 50 task-queries per cell, 10 tasks × 5 queries). Seeds: 42, 17, 1337. Best-of-3 column is the fraction of task-queries solved by at least one of the three models, averaged across seeds.

Family	Haiku 4.5	Sonnet 4.6	GPT-4o-mini	Best-of-3
`bucket_4_spatial`	0.91 ± 0.05	1.00 ± 0.00	0.74 ± 0.10	1.00 ± 0.00
`bucket_50_spatial`	1.00 ± 0.00	1.00 ± 0.00	0.64 ± 0.04	1.00 ± 0.00
`lcg_mod4_algeb`	0.24 ± 0.03	0.29 ± 0.08	0.23 ± 0.05	0.56 ± 0.03
`lcg_mod50_algeb`	0.01 ± 0.01	0.02 ± 0.02	0.03 ± 0.01	0.05 ± 0.03

Cross-seed stability. The spatial-vs-algebraic gap, averaged across the three models, is 0.629 ± 0.022 at cardinality 4 and 0.860 ± 0.014 at cardinality 50. The standard deviation across seeds is small relative to the gap magnitude on either side, indicating the boundary is a property of the rule space rather than of any specific task instantiation.

Per-model spatial-vs-algebraic gaps under matched cardinality:

Table 3: Per-model permutation tests on the spatial-vs-algebraic gap, by-task resampling, 5000 permutations.

Model	Cardinality	Observed Δ	p-value
Haiku 4.5	4	$+0.62$	$< 10^{-4}$
Haiku 4.5	50	$+1.00$	$< 10^{-4}$
Sonnet 4.6	4	$+0.66$	$< 10^{-4}$
Sonnet 4.6	50	$+0.96$	$< 10^{-4}$
GPT-4o-mini	4	$+0.62$	$< 10^{-4}$
GPT-4o-mini	50	$+0.66$	$< 10^{-4}$

Per-task variance. The gap is not driven by outlier tasks. Per-task accuracy distributions are tight: every task in `bucket_50_spatial` attains 1.00 on Haiku 4.5 and Sonnet 4.6; every task in `lcg_mod50_algeb` attains 0.00 on Haiku 4.5; the algebraic-4 distribution is uniformly low (per-task max accuracy is 0.40 for Haiku, 0.80 for Sonnet, 0.40 for GPT-4o-mini).

Hand audit. A random sample of 30 rows was verified by hand against the raw model output: 30/30 predictions match the parser output and the recomputed expected answer. Spatial-task failures occur when the model proposes a more elaborate rule (e.g. "$y = (x \bmod 17)$") instead of the obvious bucketing rule; algebraic-task failures occur when the model proposes a simpler rule (e.g. "$y = x \bmod 4$") that fits a fraction of the support but is not the true coupled rule.

5.3 Falsification of the cardinality hypothesis

The cardinality null is falsified twice. At cardinality 4, the mean spatial accuracy across three models and three seeds is 0.88 versus 0.25 algebraic ($\Delta = 0.63 \pm 0.02$). At cardinality 50, the gap *widens*: 0.88 versus 0.02 ($\Delta = 0.86 \pm 0.01$), the opposite of what cardinality alone would predict. Geometry, not cardinality, is the load-bearing variable.

5.4 Open-weight extension and the capability-emergence finding

To test whether the boundary persists outside the closed-source frontier, we run an additional sweep on two open-weight models accessed via Together.ai's serverless API: Meta's Llama 3.3 70B Instruct Turbo and Alibaba's Qwen 2.5 7B Instruct Turbo. We use the smoking-gun pair of the cardinality experiment (`bucket_4_spatial` and `lcg_mod4_algeb`) at seed 42, 50 task-queries per cell.

Boundary holds in the frontier open-weight regime. Llama 3.3 70B Turbo shows a $+0.50$ spatial-vs-algebraic gap, smaller than the closed-source frontier but of the same sign and order of magnitude. The structural pattern is robust across four vendors (Anthropic, OpenAI, Meta, Alibaba) and across the closed/open-weight divide.

Capability emergence. Qwen 2.5 7B Turbo shows a *negative* gap of -0.06: the model is at chance-level on *both* families (0.16 spatial, 0.22 algebraic). At the 7B parameter scale, the model fails uniformly. This is informative: the spatial-vs-algebraic gap is not a property of softmax attention per se. It is a property of attention architectures *above a capability threshold.*

Table 4: Open-weight extension. Cells are accuracy at seed 42 (50 task-queries each). The boundary persists in Meta's frontier open-weight 70B model. Qwen 2.5 7B is below the capability threshold for spatial recovery and shows no gap.

Model	Vendor	`bucket_4_spatial`	`lcg_mod4_algeb`	Gap
Haiku 4.5 (baseline)	Anthropic	0.84	0.22	+0.62
Sonnet 4.6 (baseline)	Anthropic	1.00	0.34	+0.66
GPT-4o-mini (baseline)	OpenAI	0.80	0.18	+0.62
Llama 3.3 70B Turbo	Meta (open-weight)	0.68	0.18	+0.50
Qwen 2.5 7B Turbo	Alibaba (open-weight)	0.16	0.22	−0.06

Below the threshold, the model is uniformly weak (no gap to observe). Above the threshold, spatial accuracy rises rapidly toward ceiling while algebraic accuracy stays at floor, and the gap emerges.

We frame this finding as: the boundary is *emergent and persistent*. It appears precisely when a model becomes capable of spatial K-shot induction and remains as that capability scales further. Characterising the threshold (the parameter scale at which the gap first appears) is open.

Excluded models. We attempted but excluded two reasoning-trained models from this sweep, DeepSeek V4 Pro and OpenAI's gpt-oss-120b. Both are "thinking-style" models that emit a separate `reasoning` field in addition to `content`. Under our $max\_tokens = 2000$ unified protocol, both models exhausted their token budget on internal reasoning and produced empty `content` fields in $\geq 30\%$ of calls. Adapting the protocol for these models requires raising max_tokens to $\geq 16{,}000$ and integrating reading of the `reasoning` field, which we leave to follow-up work characterising thinking-style models specifically. Their exclusion does not affect the present claim, which is about standard chain-of-thought ICL.

6 Best-of-N analysis

We compute observed best-of-3 ensemble accuracy and compare to the independent-failure baseline $1 - \prod_i (1 - p_i)$, where p_i is per-model accuracy on the family. The Δ between observed best-of-3 and baseline informs whether models share blind spots ($\Delta \ll 0$) or fail independently ($\Delta \approx 0$).

Table 5: Observed best-of-3 (mean over 3 seeds) vs independent-failure baseline computed at seed 42. $\Delta \approx 0$ means models fail approximately independently; the limitation is structural rather than driven by shared specific blind spots.

Family	Observed best-of-3	Independent baseline	Δ
`bucket_4_spatial`	1.00 ± 0.00	1.00	+0.00
`bucket_50_spatial`	1.00 ± 0.00	1.00	+0.00
`lcg_mod4_algeb`	0.56 ± 0.03	0.58	−0.02
`lcg_mod50_algeb`	0.05 ± 0.03	0.08	−0.03

The observation: failures are approximately independent across models (each model misses different specific queries), but per-model accuracy is uniformly low enough that pooling three independent models reaches only 0.56 on cardinality-4 algebraic and 0.05 on cardinality-50 algebraic. The boundary is structural to the rule space; pooling models does not cross it.

7 Discussion

7.1 Why does the boundary exist?

We do not provide a mechanistic proof. Spatially separable rules admit a region-by-region decomposition: a query is answered by attending to the sub-region of the support nearest to the query, ignoring the rest. Algebraically coupled rules require joint identification of multiple parameters from the full support, and the in-context attention pattern at K=16 in current frontier LLMs does not appear to yield this joint identification reliably. [1] characterizes the same boundary at the architecture level for purpose-built K-shot induction networks, where pure-attention readouts also fail on algebraically coupled families and a hybrid neural-symbolic executor crosses the boundary.

7.2 Methodology note: Sonnet 4.6 across protocols

Sonnet 4.6 spatial accuracy in the pilot is 0.24, well below the other three models (Table 1). In the cardinality test with max_*tokens* = 2000, Sonnet 4.6 reaches 1.00 spatial accuracy on both families (Table 2). Two factors plausibly contribute to the gap:

- *Output-budget truncation.* With the pilot's max_*tokens* = 15 (terse prompt), 87/150 of Sonnet's spatial responses parse as `None` because the response is cut before any integer appears. With an intermediate max_*tokens* = 500 (chain-of-thought re-run on the pilot families), 0/150 are `None` but 203/250 responses hit the 500-token cap mid-reasoning, and the parser falls back to a non-final integer. The cardinality test's max_*tokens* = 2000 avoids truncation entirely.

- *Task surface form.* The cardinality-test spatial families ($y = \lfloor x/50 \rfloor$ and $y = \lfloor x/4 \rfloor$) admit a single transparent rule. The pilot spatial families have T thresholds drawn from a coarse pool, which can leave room for spurious modular hypotheses; we observe Sonnet's chain-of-thought repeatedly proposing hypotheses such as "$y = (x \bmod 17)$" that fit a fraction of the support but are not the threshold rule.

We do not run a controlled max_*tokens* sweep on the pilot families, so we cannot fully attribute Sonnet's recovery to the budget change alone. The methodological takeaway, robust to that ambiguity, is that prompt-budget choice can confound LLM capability evaluation, and reproducibility requires fixing the protocol per task family rather than per model.

7.3 Implications

Pre-deployment capability prediction. The features listed in Section 3—input arity, output cardinality, presence of modular wrap, presence of recurrence, axis-aligned-decision-tree fit quality on the support—are computable from the task specification without invoking the LLM. These features suggest a simple offline predictor of P(success) per LLM, before any production call. We sketch such a predictor in supplementary code and leave its empirical calibration on natural agent traffic to follow-up work.

Cost-optimized routing. Spatial-separable tasks are reliably solved by all four tested models, including the cheapest. A router that classifies tasks geometrically can route the spatial fraction of a workload to the cheapest model meeting an accuracy threshold without quality loss.

Symbolic-fallback hybrids. Algebraically coupled tasks are not solved reliably by any tested model. The Relational Consistency Executor of [1] recovers polynomial coefficients at 100% via a 3×3 linear system in $\mathbb{F}_p$ and recovers LCG parameters at 96.4% via support-consistency search. A geometry-routed pipeline (LLM for spatial, RCE for algebraic) covers the union of regimes that no single component covers alone.

8 Identification vs enumeration: the priming experiment

A natural question follows from the cardinality result: *is the failure on algebraically coupled rules a failure of hypothesis-class enumeration (the model does not propose "LCG" as a candidate rule), or a failure of parameter identification (the model cannot recover* (a, c, m) *even given the rule family)?*

We test this by priming. For each `lcg_mod4_algeb` task (10 tasks × 5 queries, seed 42), we add an explicit hypothesis-class hint to the prompt:

> *The rule has the form* $y = ((a \cdot x + c) \bmod m) \bmod 4$*, where* a, c *are integers in* $[0, m)$ *and* m *is a prime in* $[503, 1009]$*. Recover the parameters* (a, c, m)*, then apply the rule to the query.*

We re-run Haiku 4.5, Sonnet 4.6, and GPT-4o-mini under this primed protocol with otherwise identical settings ($\max\_tokens = 2000$, same FINAL ANSWER format).

Table 6: Priming experiment. Accuracy on `lcg_mod4_algeb` (seed 42, 50 task-queries per cell) with vs without the explicit hypothesis-class hint.

Model	Baseline (no prime)	Primed	Δ
Haiku 4.5	0.22	0.24	+0.02
Sonnet 4.6	0.34	0.22	−0.12
GPT-4o-mini	0.18	0.32	+0.14
Mean	0.25	0.26	+0.01

The priming hint produces no meaningful improvement. The mean primed accuracy (0.26) is statistically indistinguishable from the unprimed baseline (0.25). Per-model deltas range from −0.12 to +0.14, none of which is significant under the same by-task permutation procedure used in Section 5.

Interpretation. The failure on algebraically coupled K=16 induction is *not* a failure to enumerate the correct hypothesis class; even when the rule family is given explicitly with parameter-range bounds, the models cannot recover the parameters from 16 examples. The bottleneck is parameter identification — jointly solving for (a, c, m) over the K in-context constraints. We conjecture that softmax attention with $K = O(1)$ tokens of context implements an interpolation-style operation that does not natively support the joint constraint satisfaction required for parameter identification under modular arithmetic. The priming result is consistent with both the architectural-bottleneck hypothesis (Section 7) and the pre-training-distribution hypothesis (no algebraic patterns implicitly encoded), and we propose follow-up experiments to disambiguate these.

9 Limitations

- $K = 16$ is one specific in-context budget. We additionally tested $K \in \{64, 256\}$ on the smoking-gun pair and observed that the algebraic accuracy stays at ~ 0.24 while spatial saturates at 1.00; the gap widens with K. We do not characterize $K < 16$.

- Five operator families is a small taxonomy. String transduction, graph traversal, and stateful planning are not covered.
- Reasoning-trained "thinking-style" models (DeepSeek V4 Pro, gpt-oss-120b) are excluded due to protocol incompatibility (see Section 5.4). They require a different harness (large max_*tokens*, separate handling of the `reasoning` field) and are deferred to follow-up work.
- The capability threshold below which the gap disappears (Section 5.4, Qwen 2.5 7B) is not characterized. At what model scale does spatial saturation precede algebraic ceiling?
- Anthropic Claude Opus 4.7 is included in the pilot but omitted from the cardinality test for cost. Its qualitative behavior in Table 1 matches Haiku and supports the same conclusion.
- We do not provide a mechanistic explanation, only a structural one. The mechanism that distinguishes spatial recovery from algebraic recovery in transformer in-context computation is open.
- Frontier LLM training data may have leaked specific algebraic patterns. We use prime $m \in [503, 1009]$ outside common pedagogical primes to mitigate, but cannot fully rule out contamination.

10 Conclusion

When a rule's parameters can be locally recovered from regions of the input support, commercial frontier-family LLMs induce it well from K=16 examples in-context. When the parameters are algebraically coupled and require joint identification across the full support, the same models fail—even under cardinality control, even pooled into a three-model ensemble, even when the rule has only 4 output classes. The boundary survives at $p < 10^{-4}$ on all six per-model contrasts and is stable across three independent task-suite seeds. We propose this as the empirical foundation for pre-deployment LLM capability prediction and a routing principle for hybrid LLM-symbolic pipelines.

Reproduction. Code, tasks, raw per-call results, and the predictor artifact are released at `https://github.com/`*[anon]*`/llm-geometric-boundary`. Persistent identifier: `10.5281/zenodo.[pending]`.

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A Pilot suite specification

A.1 Operator families

- `threshold`: $y = \mathbb{1}[x > T]$, T drawn uniformly from $\{15, 23, 38, 47, 62, 71, 89, 113, 127, 151\}$, $x \in [0, 200]$.
- `conjunctive`: $z = \mathbb{1}[x > T_1 \wedge y > T_2]$, both T drawn uniformly from $\{15, 30, 45, 60, 80, 100, 120, 140\}$, $(x, y) \in [0, 180]^2$.
- `decision_tree`: depth-2 axis-aligned tree with 3 thresholds T_1, T_2, T_3 and a permutation of $\{0, 1, 2, 3\}$ as leaves; rejection sampling enforces all 4 leaves appear in the $K = 16$ examples.
- `lcg`: $y_t = (ax_t + c) \bmod m$, $m \in \{107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173, 179, 181, 191,$ (excluding 103 for sampling). $K = 16$ consecutive states observed.
- `polycoef`: $y = (ax^2 + bx + c) \bmod p$, $p \in \{11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83,$

A.2 Pilot prompt protocol

The pilot uses *terse* system messages instructing the model to "output ONLY the integer" with $max\_tokens = 15$ for Haiku, Opus, and GPT-4o-mini, and the chain-of-thought protocol of Section 3.6 for Sonnet 4.6 (a re-run after observing parsing failures with the terse protocol). This mixed methodology is the source of the protocol-heterogeneity caveat in Section 4 and is rectified in the unified protocol of Section 5.

B Cardinality suite specification

All four cardinality families share 1-D integer inputs $x \in [0, 199]$, $K = 16$ examples, 5 heldout queries, no x overlap between examples and queries.

- `bucket_4_spatial`: $y = \lfloor x/50 \rfloor$. Rejection sampling enforces all 4 output classes appear.
- `bucket_50_spatial`: $y = \lfloor x/4 \rfloor$. With $K = 16$ examples uniformly drawn from $[0, 199]$, between 12 and 16 of the 50 output classes are observed.
- `lcg_mod4_algeb`: $y = ((ax + c) \bmod m) \bmod 4$, with m drawn uniformly from a pool of 30 primes in $[503, 1009]$, and $a \in [2, m - 1]$, $c \in [0, m - 1]$ random. Rejection sampling enforces all 4 output classes appear in the $K = 16$ examples.
- `lcg_mod50_algeb`: same as above with $\bmod 50$ reduction.

C Hand audit (sample)

A random sample of 30 rows from the cardinality experiment was inspected by hand. For each row we recorded the recomputed expected answer, the parsed predicted answer, and the trailing 250 characters of the raw response. All 30 rows are consistent (parsed answer matches raw output and recomputed expected) with 0 parser bugs. Sample failures show the predicted failure mode of each geometry: for spatial rules, occasional over-engineering (model proposes a modular rule when bucketing was the truth); for algebraic rules, systematic underfitting (model proposes a simpler heuristic that explains a fraction of the support).